Panos Antonopoulos

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SUMMARY

Data scientist and physicist with a Distinction in MPhil Data Intensive Science from the University of Cambridge and a First-Class Honours BSc in Physics from the University of Edinburgh, graduating top of cohort. Skilled in machine learning, AI, and high-performance computing, with strong experience in Python, C++, and data analysis. Recognised with multiple academic awards, including the Neil Arnott Prize and Class Medals, for consistently ranking first in year. Developed leadership and communication skills through research, tutoring, and participation in the Cambridge Future Global Leaders Programme.

EDUCATION

UNIVERSITY OF CAMBRIDGE — Cambridge, UK
MPhil in Data Intensive Science, Distinction (78%) — Oct 2024 – Jul 2025

Dissertation: Using Satellite Multispectral Data to Monitor Vegetation Trends as a Result of Climate Change

Relevant Courses: Statistics, Machine Learning, Research Computing and Software Development, Deep Learning, High Performance Computing, Medical Imaging, Image Analysis

THE UNIVERSITY OF EDINBURGH — Edinburgh, UK
BSc in Physics, First Class Honours (83%) — Sep 2020 – May 2024

Ranked top of cohort for overall degree

Dissertation: Investigating the Multiplicity of Neutron Interactions in the LUX-ZEPLIN Dark Matter Detector

Relevant Courses: Electromagnetism, Quantum Mechanics, Thermal Physics, Condensed Matter Physics, Relativity, Nuclear and Particle Physics, Medical Physics, Fourier Analysis and Statistics, Dynamics and Vector Calculus, Linear Algebra and Several Variable Calculus, Modelling and Visualisation in Physics, Data Acquisition and Handling, Scientific Image Analysis, Numerical Recipes, Computer Simulation

AWARDS & ACHIEVEMENTS

UNIVERSITY OF CAMBRIDGE
- Co-author of an abstract submitted to the AGU 2025 Conference, based on master's thesis research

THE UNIVERSITY OF EDINBURGH
- Neil Arnott Prize for ranking top of cohort in Year 4
- Year 4 Physics Senior Honours Class Medal
- Neil Arnott Scholarship for ranking top of cohort in Year 3
- Year 3 Physics Junior Honours Class Medal
- Year 1 & 2 Pre-Honours Certificates of Merit

WORK EXPERIENCE

SELF EMPLOYED — Edinburgh, UK
Mathematics & Science Tutor — Jun 2021 – Jul 2024

- Guided students through complex concepts and exam preparation, improving comprehension and performance

UNIVERSITY PROJECTS

LOW-RANK ADAPTATION APPLIED TO LARGE LANGUAGE MODELS Mar 2025

- Investigated the performance of the Qwen2.5-0.5B-Instruct model in forecasting a predator prey time series modelled by the Lotka-Volterra equations
- Fine-tuned the model using the low-rank adaptation technique and compared performance

OPTIMISING AND PARALLELISING CODE TO SIMULATE HEAT DIFFUSION Mar 2025

- Optimised C++ simulation code using techniques such as loop unrolling, pointer manipulation, data structure refinement, and code hoisting, achieving a 3.5x performance improvement
- Parallelised the simulation with MPI (Message Passing Interface), delivering an additional 1.8x speedup on top of single-thread optimisation
- Developed the code to be compatible with supercomputers

PROCESSING PET-CT AND MRI MEDICAL IMAGES Mar 2025

- Reconstructed PET-CT scans of lungs to create an interpretable image
- Denoised MRI images of a knee and combing images from different coils
- Developed an algorithm to segment a tumour in a CT scan

DEVELOPING A PYTHON PACKAGE TO PERFORM AUTOMATIC DIFFERENTIATION Dec 2024

- Created a fully functional Python package to perform automatic differentiation using dual numbers
- Cythonised the package to improve performance by up to 2x
- Wrote tests, documentation, and a tutorial on how to use the package

EXTRACURRICULAR ACTIVITIES

LUCY CAVENDISH COLLEGE, UNIVERSITY OF CAMBRIDGE Cambridge, UK
Future Global Leaders Programme Oct 2024 – Jul 2025

- Strengthened leadership, critical thinking, and problem-solving skills by working on a UN Sustainable Development Goal project, earning second place with team project
- Learning how academic studies apply outside academia through visits to Arm and AstraZeneca

SKILLS

Technical Skills: Python, C++, PyTorch, Matplotlib, SciPy, Keras, Pandas, Skimage, Git, Seaborn, TensorFlow, Deep Learning, Machine Learning, Computer Vision, Linux, Jupyter Notebook, Artificial Intelligence, Data Science, Scikit-Learn, Convolutional Neural Networks, Data Visualisation, Anaconda, Data Analysis, NumPy, Object-Oriented Programming, Data Preprocessing, LaTeX, Software Development, Docker, Natural Language Processing, Microsoft Office, Microsoft Word, Microsoft Excel, Microsoft PowerPoint, Google Drive

Soft Skills: Problem Solving, Report Writing, Presenting, Time Management, Teamwork, Leadership

ADDITIONAL

Certifications: J.P. Morgan Investment Banking Virtual Experience Program Participant (Forage), Goldman Sachs Engineering Virtual Experience Program Participant (Forage)

Hobbies: Researching and investing in the equity market, keeping up to date with global affairs, politics and international relations, running, swimming, hiking and dog walking